Tile Editor Modding Guide

Authoring map mods, data formats, and runtime components

Contents

Data Formats And Examples

Everything the Tile Editor authors is written as portable JSON inside a map mod folder. This page shows each format with a working example and explains what the game does with it.

The important design point: **the runtimes that consume these files do not require the Tile Editor.** Players install the small runtime mod; you ship the JSON inside your map mod.

File	Consumed by	Purpose
`game-graph.json`	RailLoader / FUSE	Track nodes and segments
`grade-crossings.json`	`Hrogers.CrossingRuntime`	Functional grade crossings
`train-signals.json`	Railroad Operations `Hrogers.SignalRuntime`	Semaphore signals and interlockings
`tile-editor-telegraph-poles.json`	Tile Editor bridge	Pole nodes and wire edges
`Map.json`	Tile Editor	Georeference for the tile set
`Definition.json`	RailLoader	Mod manifest (version 8)

Grade Crossings

`grade-crossings.json` registers native `TrackMarkerType.Crossing` markers. Railroader's normal Auto Engineer crossing setting then controls bell and horn behaviour. Because the markers live in the shared track graph, they work for player-owned equipment in Waypoint mode as well as AI equipment.

Minimal

```
{
  "version": 1,
  "crossings": [
    {
      "id": "bryson-main-street",
      "enabled": true,
      "nodeId": "NBrysonCrossing"
    }
  ]
}
```

With no `segmentIds`, **every segment connected to the node** receives a marker, so both approaches detect the crossing. That is what you want almost always.

Limiting the protected approaches

At an unusual junction where only some approaches should sound, name them:

```
{
  "version": 1,
  "crossings": [
    {
      "id": "yard-throat-crossing",
      "enabled": true,
      "nodeId": "NYardThroat",
      "segmentIds": ["SMainWest", "SMainEast"]
    }
  ]
}
```

Only `SMainWest` and `SMainEast` get markers; the yard leads off the same node stay silent.

In use

1. In the editor, select the track node where the road crosses.
2. Author the crossing so it writes into your map mod's `grade-crossings.json`.
3. Ship `Hrogers.CrossingRuntime` alongside the map, or tell players to install it.
4. In game, run a train toward the node with Auto Engineer crossing behaviour on — the bell and horn fire on approach.

Set `"enabled": false` to keep a crossing in the file but inactive, which is better than deleting it while you test.

Train Signals

`train-signals.json` places Railroader's own animated semaphore assemblies. It is loaded during ordinary gameplay by Railroad Operations — players do not install the Tile Editor.

A single mast

```
{
  "formatVersion": 1,
  "signals": [
    {
      "id": "sig-bryson-west",
      "enabled": true,
      "headCount": 2,
      "initialAspect": "clear",
      "position": { "x": 1240.5, "y": 312.0, "z": -880.25 },
      "rotation": { "x": 0.0, "y": 137.5, "z": 0.0 },
      "protectedNodeId": "NBrysonWest",
      "protectedSegmentId": "SMainWest",
      "direction": "forward"
    }
  ]
}
```

`headCount` accepts one, two, or three heads. `initialAspect` is one of `clear`, `restricting`, or `stop`. `direction` is `forward` or `reverse` relative to the protected segment.

Locking a mast to the track

A mast can either be snapped to the rail once, or **locked** to it. A locked mast keeps its side, height, and facing offsets and follows later Bezier curve, grade, and elevation edits:

```
{
  "id": "sig-grade-approach",
  "enabled": true,
  "headCount": 1,
  "initialAspect": "restricting",
  "trackAttachment": {
    "locked": true,
    "segmentId": "SMainWest",
    "parameter": 0.62,
    "localPosition": { "x": 3.4, "y": 0.0, "z": 0.0 },
    "localRotation": { "x": 0.0, "y": 0.0, "z": 0.0 }
  },
  "protectedSegmentId": "SMainWest",
  "direction": "forward"
}
```

`parameter` is the position along the segment from 0 to 1. Lock masts you expect to survive regrading; use a one-time snap for scenery-only signals.

Diamond interlocking

The Diamond Interlocking builder detects the real intersection of two non-connecting Bezier segments, rejects a grade-separated overpass, and places four independently adjustable masts — `A1`, `A2`, `B1`, `B2`:

```json
{
  "formatVersion": 1,
  "interlockings": [
    {
      "interlockingId": "dia-hollow-junction",
      "type": "diamond",
      "automatic": true,
      "crossingPoint": { "x": 980.0, "y": 244.5, "z": 1502.0 },
      "crossingAngleDegrees": 63.4,
      "releaseLength": 120.0,
      "releaseDelaySeconds": 8.0,
      "cancelDelaySeconds": 30.0,
      "approaches": [
        {
          "approachId": "A1",
          "signalId": "sig-dia-a1",
          "approachLength": 650.0,
          "protectedSegmentIds": ["SRouteA-01", "SRouteA-02"],
          "approachSegmentIds": ["SRouteA-03", "SRouteA-04"],
          "conflicts": ["B1", "B2"]
        },
        {
          "approachId": "B1",
          "signalId": "sig-dia-b1",
          "approachLength": 650.0,
          "protectedSegmentIds": ["SRouteB-01"],
          "approachSegmentIds": ["SRouteB-02", "SRouteB-03"],
          "conflicts": ["A1", "A2"]
        }
      ]
    }
  ]
}
```

Two things to note:

- `protectedSegmentIds` **and** `approachSegmentIds` **are different.** The first is the block from mast to diamond; the second is the approach-locking territory beyond the mast. Exposing them separately is what lets external interlocking logic reason about the plant.
- **Long setbacks follow connected track automatically.** A 500–800 m setback crosses segment boundaries without you listing every segment by hand — the arrays above record the result.

`conflicts` names the approaches this one locks out, which is what makes the diamond mutually exclusive.

Telegraph Poles

New poles and wire edges persist in the owning map mod's `tile-editor-telegraph-poles.json`. Original base-game poles instead save cumulative Alina `TelegraphPoleMover` offsets plus portable rotation overrides, so both kinds coexist.

```
{
  "version": 1,
  "poles": [
    { "id": "pole-001", "position": { "x": 1200.0, "y": 310.5, "z": -900.0 },
      "rotation": { "x": 0.0, "y": 42.0, "z": 0.0 } },
    { "id": "pole-002", "position": { "x": 1260.0, "y": 311.2, "z": -880.0 },
      "rotation": { "x": 0.0, "y": 42.0, "z": 0.0 } }
  ],
  "edges": [
    { "from": "pole-001", "to": "pole-002" }
  ]
}
```

Poles are nodes; `edges` are the wire runs between them. Continuous-line placement creates the poles and the edges in one pass.

Track Graph And Gauge

`game-graph.json` stays the portable baseline. FUSE imports it including Narrow Gauge metadata, and RailLoader safely ignores the extra `gauge` field — which is why one file serves both runtimes.

```
{
  "segments": {
    "SMainWest": {
      "startId": "NBrysonWest",
      "endId": "NBrysonYard",
      "trackClass": "Mainline",
      "style": "Standard",
      "speedLimit": 45,
      "priority": 0,
      "groupId": "bryson-main",
      "gauge": "DualGauge_L"
    }
  }
}
```

Gauge values: `Standard` (or omitted), `Narrow`, `DualGauge`, `DualGauge_L`, `DualGauge_R`, `DualGauge_T`. Visible narrow and dual rail geometry comes from FUSE Narrow Gauge, not Railroader's standard track builder — the field here is metadata that module reads.

`DualGauge_T` is authored as **one short segment** between opposite explicit `L` and `R` runs. The in-game F9 workspace checks its two endpoints and refuses to apply it through a whole chain, which is the mistake it exists to prevent.

Native FUSE Packages

Native FUSE packages are editable directly. The desktop and F9 selectors read `Info.json`'s `FuseDataFiles`, and `.fuse.json` track fragments keep `startNodeId` / `endNodeId`, native removal lists, and unrelated FUSE operations rather than being rewritten as legacy JSON.

Use RailLoader `Definition.json` (manifest version 8) for portable content, and native FUSE fragments when the legacy schema cannot express what you need.

Choosing A Format

You want	Author as
Track that works on RailLoader and FUSE	`game-graph.json`
Narrow / dual gauge	`game-graph.json` with `gauge` (needs FUSE Narrow Gauge to render)
FUSE-only operations	native `.fuse.json` fragment
Crossings for any player	`grade-crossings.json` + `Hrogers.CrossingRuntime`
Signals for any player	`train-signals.json` + Railroad Operations

Related

- Track Editing
- Mod Tools
- FUSE JSON Schema

Mod Tools

The second nav row. These panels edit everything in a map mod that isn't terrain or raw track geometry.

Every panel writes to the owning mod JSON layer and triggers a bridge reload. `Ctrl+Z` undoes panel edits too, up to 50 steps.

Mod Panel

Control	What it does
Open Mod Folder	Load a mod folder (needs `Definition.json`)
Open Base Game	Load a standalone `graph-data.json`
New Mod	Create a new mod folder structure
Save All	Save every dirty layer file
Layer dot	Toggle layer visibility
Layer row	Set as the active layer
● indicator	Unsaved changes in that layer

Columns show per-layer counts: Nodes (plus deleted), Segs, Spl (splineys), Areas.

Closing the panel with ✕ or `Esc` leaves the mod loaded.

Map Legend

What you see with a mod loaded:

Appearance	Meaning
Yellow lines	`game-graph.json` track segments
Grey lines	Base game track
Blue polylines	Rivers (FlowyThingBuilder)
Tan polylines	Roads (FlowyThingBuilder)
Grey/cream	AutoTrestle bridge spans
Orange circles	Turntables
Green diamonds	Stations
Orange diamonds	Loaders / industry
Coloured squares	Scenery buildings — zoom in for the model name
Large circles	Area/town centres with name labels
Orange circle	Live locomotive (from the bridge)
Teal square	Live railcar (from the bridge)

The last two only appear with a live game connected.

Progression Editor (Prog)

Left column lists sections in topological order, prerequisites first. Right column lists map features and what each unlocks.

Control	What it does
+ Section	Add a purchasable section
+ Feature	Add a map feature
Del Section / Feature	Remove the selected item
Save	Write back to `progressions.json`

Section fields: id, display name, prerequisites, cost, feature to enable. Feature fields: id, display name, areas unlocked, track groups.

Topological ordering means a section always appears after everything it depends on — if one is in an unexpected place, its prerequisites are the reason.

Area / Town Editor (Areas)

Three columns: areas (sorted by order, with a source-layer colour dot), industries in the selected town, and components in the selected industry.

Control	What it does
Click a row	Select and load the next column
Go to Area	Pan the map there and close the panel
+ Area	Create at the current map centre
+ Industry	Create in the selected town
+ Component	Create in the selected industry
Edit Area / Industry / Comp	Edit that JSON object directly
Del Area	**Mark** for deletion (not saved yet)
Del Industry / Comp	Remove immediately from its parent
Save	Write all dirty town JSON files

Note the asymmetry: deleting an area only marks it, while deleting an industry or component removes it from the parent right away.

Industry creation exposes the full RailLoader/FUSE component set — storage capacity and daily change, car transfer rates and ordering, formula input/output maps, team-track import/export profiles, passenger population and neighbours, repair/overhaul, interchange variants, progression, multiple TrackSpans, and custom typed fields.

Scenery Placement (Scenery)

Control	What it does
Model ID field	Type an identifier or click a quick-pick
RotY nudge	±90 / 45 before placing

Control	What it does
Place on Map	Placement mode — click to drop
`Esc`	Exit placement mode
Go To	Pan to the selected object
Del Object	Remove from the layer

Y position is auto-sampled from terrain at the click point.

The in-game Scenery workspace adds a searchable loaded-asset palette that includes runtime-resolvable RailLoader `SCAssetPacks` registered after the initial catalog, plus in-world picking, transform controls, terrain snapping, and duplication.

Mandela / Prefab Instances (Mandela)

Places instances of base-game prefabs.

Control	What it does
Target	Destination scene path key
Prefab	Base-game prefab path to instantiate
Base Pick	Search `dumped-mandelas.txt` and fill Prefab
Reload Base	Reload the dumped prefab catalog
Load Sel / Save Sel	Pull draft values from an entry, or write them back
Duplicate	Copy a placement to a new target path
Place on Map	Drop the draft at terrain height
Enabled	Cycle default / enabled / disabled

FUSE imports the same entries as `world.sceneClones`, so one graph stays compatible with both runtimes.

Objects Workspace (in-game)

Selects base-game buildings and props under the mouse and saves move, rotate, scale, enable/disable, and safe-clone operations as portable RailLoader mandelas.

Shared town, map, and world roots are blocked, so clicking one station cannot move the entire scene. Thin signs and small props get a screen-space picking halo when the ray misses.

The F9 OBJECTS page also identifies Railroader's original scene signs and gives them a dedicated visible/hidden toggle, without treating loader-added scenery signs as base-game objects.

Track Spans (Spans)

Lists all spans across layers with their layer colour. Click a span to edit its upper and lower segment, distance, and end. **+ New Span** and **Del** manage entries.

Spans may cover a whole segment, a measured partial range, or connected start/end segments.

Group Move (Group)

`Ctrl+drag` rubber-bands a node selection; `Shift`+rubber-band adds to it. Apply dX / dY / dZ / Rot to translate and rotate everything selected at once.

Calculators (Calc)

Calculator	Input → output
Crossover	Separation + angle → leg lengths
Curved Turnout	Radius + gauge + angle → diverge geometry
Grade / Slope	Run + rise → percent, ratio, angle
Measure	Run, rise, grade, heading between picks

Spliney Panel

Control	What it does
Type	Road or River
Target	An existing writable spliney layer
New JSON…	Create and register a new road or river file
Width / Seed	Defaults for new placements
Use Sel Rot	Use the selected track node's heading
Place on Map	First click creates a spliney with two starter points
Go To / Delete	Navigate to or remove the selected spliney

The details card shows style, width, point count, and length.

Loaded base-game roads and rivers also expose their real control nodes. The first change creates a same-name override in the selected graph, and every adjustment refreshes the associated height, roadbed, dirt, object, water, and vegetation-mask terrain modifiers.

Telegraph Poles (in-game)

Amber world markers, pole creation at the pointer, continuous-line placement, manual wire connections, WORLD/LOCAL movement, full pitch/heading/roll rotation, and deletion.

New poles and edges persist in the map mod's `tile-editor-telegraph-poles.json`. Original poles still save compatible cumulative Alina `TelegraphPoleMover` offsets plus portable rotation overrides.

Related

- Data Formats — what these panels write
- In-Game Geo Workspace
- Track Editing

Track Editing

Building and editing track graphs on real terrain.

Every change auto-saves to `game-graph.json` and hot-reloads into a running game in roughly 500 ms.

Selecting

Action	Result
Click a node or segment	Select — properties appear top-left
Click a segment	Works **anywhere along the line**, not just the midpoint
Click empty space	Deselect
Click an already-selected node	Second click starts drag mode
Release a drag	Commits the move and samples terrain height
`Esc`	Cancel drag, connect, or place
`Delete`	Delete the selected node, or all its segments

The two-click drag is deliberate: one click selects, a second starts the move, so you cannot nudge a node by accident while inspecting it.

Node Properties

Control	What it does
X / Y / Z	Direct position editing — click, type, Enter
RotY nudge	Fine grid: ±90 / 45 / 30 / 15 / 10 / 5 / 1 / 0.1 / 0.05 / 0.001°
Flatten	Zeros rotX and rotZ, levelling the node on terrain
Reverse	Flips heading 180°
Merge	Removes a middle node between exactly two segments
Split	Duplicates the node and rewires the selected segment to the copy
Copy XYZ	Copies position to the clipboard
Paste Y	Pastes **only the height** onto the node

Paste Y is the tool for levelling a run: copy from a node at the right elevation, then paste height onto each node along the tangent without disturbing plan position.

The connected-segments list at the bottom of the node panel is clickable.

Segment Properties

Control	What it does
Track Class	Mainline / Branch / Industrial — coloured buttons
Style	Standard / Yard / Bridge / Tunnel
Speed	Nudge ±25 / 10 / 5 / 1 mph, or type a value

Control	What it does
Priority	Editable box
GroupID	Track group membership
Reverse	Swaps `startId` and `endId`
Trestle	Wraps the segment in an AutoTrestleBuilder spliney

A class button brightens when that class is active.

Creating Track

Action	Result
`Ctrl+click` map	Create a node at the cursor, sampling terrain Y
Geo → Add Node	Placement mode with a live-coordinate crosshair
Connect → button	Enter connect mode from the selected node
`Ctrl+click` a node	Finish the connection
Drag node → node	Snap ring; release to connect
Drag node → segment	**Cyan**; release to insert into that segment
`Shift+drag` → segment	**Yellow**; insert **and** add a turnout diverge leg

New nodes default to Mainline, Standard, 45 mph, and their `rotY` is set automatically to face the connection direction.

Inserting Into A Segment

Dropping a node onto a segment deletes the original and creates two, giving the new node a bezier heading so the alignment stays smooth. The `Shift` variant does the same and adds a turnout diverge leg using the current **Geo** → **Turnout** settings for angle and length.

For a full switch with proper geometry, use **Geo** → **Turnout**: select the node, Preview, then Commit.

Group Operations

Action	Result
`Ctrl+drag`	Rubber-band select nodes
`Shift`+rubber-band	Add to the existing selection
Apply dX / dY / dZ / Rot	Bulk translate and rotate the selection

Spliney Control Points

Rivers, roads, and trestles are splines with control points — the yellow dots. Zoom in to click one.

Control	What it does
Click, click again	Select, then start dragging

Control	What it does
Drag release	Moves the point; terrain can resample Y
Prev / Next	Step along the spline
Ins Before / Ins After	Insert a control point
Sample Y	Re-read terrain height
Auto Rot	Rebuild `rotY` from neighbouring points
Delete Pt	Remove a point (the spline keeps at least two)
Fit Terrain	Terrain-fit the entire road or river
Reverse Flow	**Rivers only** — reverses points and adds 180° `rotY`

River preview arrows show flow from the first point to the last, so check them after a Reverse Flow.

Gauge

Segments carry a `gauge` field consumed by FUSE Narrow Gauge:

`Standard` (default) · `Narrow` · `DualGauge` · `DualGauge_L` · `DualGauge_R` · `DualGauge_T`

New arcs, pieces, parallel tracks, turnouts, wyes, and direct connections inherit the active gauge. Existing gauge and companion fields survive splits, merges, and renames.

RailLoader ignores the `gauge` field, so one graph works for both runtimes. Visible narrow and dual rail geometry is rendered by FUSE Narrow Gauge, not by the Tile Editor — the field is metadata.

`DualGauge_T` must be **one short segment** between opposite explicit `L` and `R` runs. See Data Formats.

Geometry Tools (Geo)

The Geo panel carries seven tabs — Spliney, Arc, Parallel, Fit Arc, Turnout, Wye, Span, and Turntable, alongside grade work and piece-based assembly.

Guide geometry stays draft-only until you fit an arc or build a spliney, so you can trace freely without committing anything.

Calculators (Calc)

Calculator	Input → output
Crossover	Separation + angle → leg lengths
Curved Turnout	Radius + gauge + angle → diverge geometry
Grade / Slope	Run + rise → percent, ratio, angle
Measure	Run, rise, grade, and heading between two picks

Related

- Keybinds
- Mod Tools
- Data Formats

In-Game Geo Workspace

`Hrogers.TileEditorBridge` puts a live editing workspace inside Railroader, on top of the game's own `Track.Graph`. Press **F9**.

It does not use Alina's Map Editor, and it can select and remember an installed RailLoader mod or game-graph by itself — **the desktop Tile Editor does not need to be running.**

Mouse Handling

While F9 is open the mouse is reserved for editing. Railroader's pan, orbit, look, and wheel zoom are locked so the camera stays put during placement.

- `W A S D` and the normal fast-movement modifiers still move the camera.
- **Hold middle mouse** to temporarily restore all normal camera controls. Editing clicks are ignored until you release it.
- **Right-click the world** to clear every selection and cancel any active mode — placement, connection, bridge-picking, Fit Arc, TrackSpan, pole-wiring, terrain, and delete.

That right-click is the universal escape hatch; reach for it when a mode seems stuck.

Workspaces

Workspace	Covers
Geo	Nodes, segments, grades, pieces, arcs, parallel track, fitted arcs, turnouts, wyes, splineys
Terrain	Sculpting, plus vegetation and water painting, and the OSM guide
Scenery	Searchable asset palette, placement, transforms, terrain snapping
Objects	Base-game buildings and props → portable mandelas
Poles	Telegraph pole nodes and wire edges
Signals	Semaphore placement and interlockings
Operations	Towns, spans, industries, stations, loaders, turntables
Desktop	Editor status; remotely focus, undo, save, and reload

The panel is resizable.

Track Editing

A fast shortcut for connected track: **click the first node normally, then Shift-click the second** to connect them. The destination stays selected, so you can keep Shift-clicking to build a chain.

Action	Result
Place Free Node	Create and select an independent starting node anywhere on terrain
Add +10 m	Create a connected node ahead using current heading, grade, and bank
`Ctrl`+drag a node	Move it over terrain as one undoable edit
Drop over a cyan node	Connect the two, keeping the dragged node at its last valid position

Repeated **Add +10 m** clicks extend track continuously, selecting each new endpoint — the quickest way to run a tangent.

Node transforms live in their own movable, resizable child window. The main Geo surface keeps a compact summary and continue-track controls, while naming, move/rotate, exact coordinates, actions, and selective copy/paste stay together in the focused Node Editor.

Its direction pad separates elevation from WORLD/LOCAL plan movement and groups pitch, heading, and roll into paired curved-arrow controls, with hover text on every axis.

Property Clipboard

Copy a single field rather than the whole node: elevation, grade, heading, bank, full rotation, an elevation+rotation combination, the turnout switch flag, or everything. Only compatible paste actions light up on the target. X/Z position and connections are never changed.

Naming

New nodes can use a remembered prefix and readable name pattern. Every builder shares the pattern and adds a collision-safe number.

Gauge

Track creation and segment properties support `Standard`, three-foot `Narrow`, automatic `DualGauge`, explicit `DualGauge_L` / `DualGauge_R`, and `DualGauge_T` transitions. Track class can be changed live between Mainline, Branch, and Industrial with undo/redo and schema-safe output.

F9 distinguishes saved gauge metadata from a loaded live runtime, tells you when FUSE and FUSE Narrow Gauge must be enabled before restarting, and offers a live gauge synchronisation action when both are active.

Performance detail worth knowing: ordinary 3-foot node and segment edits batch their FUSE metadata and rebuild only affected endpoints. **Dual-gauge topology edits are deferred and coalesced**, because their generated ghost and shared rails need the complete synchroniser — so a dual-gauge change costs more than a narrow one.

`DualGauge_T` is authored as one short segment between opposite explicit L/R runs; F9 checks its two endpoints and prevents applying it through a whole chain.

Signals

Places Railroader's base-game animated semaphore assemblies with one, two, or three heads: pointer placement, amber world selection, WORLD/LOCAL movement, full rotation and flip controls, exact transforms, and test aspects.

Masts support a one-time rail snap or a **persistent track lock**. A locked mast keeps its side, height, and facing offsets and follows later Bezier curve, grade, and elevation edits.

The **Diamond Interlocking** builder detects the real intersection of two non-connecting Bezier segments, rejects a grade-separated overpass, and places four independently adjustable A1/A2/B1/B2 semaphores. Setback, lateral and vertical offset, head count, approach locking, and release length are all configurable. Long 500–800 m setbacks follow connected track across segment boundaries automatically.

Output is portable `train-signals.json`, loaded during ordinary gameplay by Railroad Operations — players never install the Tile Editor. See Data Formats.

Live **Signals & CTC**, **Train Orders**, and **My Orders** desks join the normal Company → Operations window. F9 remains the map-authoring editor.

Splineys And Bridges

The in-game Spliney workspace can build a bridge directly from a clicked track segment. Its two endpoint controls inherit the track node positions and rotations, reproducing Railroader's exact 3D Bezier span with an adjustable below-rail deck offset and no redundant intermediate nodes. Yellow track picking stays armed for the next bridge afterwards.

Road, river, bridge, and trestle control points show movement plus full pitch/heading/roll together, using the same arrow controls and precision steps as track nodes and scenery.

Operations

Discovers, searches, selects, and edits towns, TrackSpans, industries, passenger stops, freight components, engine facilities, physical coal/water/fuel loaders, custom loader prefabs, station agents, commodities, and turntables. Pointer placement and coloured world overlays use the same undoable document workflow as track and scenery.

Dedicated Geo **Span** and **Turntable** tools sit beside Arc, Turnout, and Wye. Turntables support native FUSE pit and bridge geometry, bridge-track gauge, subdivisions, and optional roundhouse stalls, plus preserved legacy RailLoader/Alina `TurntableBuilder` output. Standard 30 m and three-foot narrow-gauge presets are included.

Sync With The Desktop Editor

Track, scenery, road/river/bridge splineys, telegraph poles, and terrain tiles synchronise in **both directions**.

Dirty-edit ownership locks prevent simultaneous writes. An incoming reload preserves unsaved work as a timestamped conflict copy rather than discarding it.

`Mods/TrackBridge` files carry the live graph and reload traffic; the compact UMM panel adds a small status and request channel beside them without changing that format.

Rendering Notes

Yellow segment overlays live in a protected graph-level editor layer keyed by segment id, so Railroader track rebuilds and undo/redo no longer permanently remove editable track lines.

Overlays share lightweight materials, cache repeated asset searches, reduce long-segment pick markers, and sleep distant track visuals and colliders until the camera approaches.

Related

- Data Formats
- Mod Tools
- Getting Started

Terrain Editing

Sculpting heightmaps, painting vegetation and water, and generating tiles.

Press `E` to toggle edit mode. The cursor becomes a brush and a toolbar appears below the nav bar. **LMB paints, RMB erases.**

Brushes

Cycle with `B` in edit mode.

Brush	Effect
Raise	Raise (LMB) or lower (RMB)
Flatten	Level the area to the height you first clicked
Paint	Stamp an exact target height
Smooth	Average pixels with their neighbours
Noise	Add fBm procedural noise
Erode	Simulate hydraulic erosion

Flatten samples on first click, so click on the elevation you want *before* dragging across the area you want levelled.

Brush Controls

Key	Action
`[` / `]`	Size down / up
`Ctrl+Scroll`	Size
`-` / `=`	Strength down / up
`MMB`	Eyedropper — sample height or vegetation at the cursor

Height Clamps

Clamps bound what a brush can do, which is how you grade against a fixed elevation without overshooting.

Key	Action
`,`	Set floor clamp to cursor height
`.`	Set ceiling clamp to cursor height
`Ctrl+,`	Clear floor clamp
`Ctrl+.`	Clear ceiling clamp

Set a floor at rail height and Smooth aggressively around a right-of-way — the clamp stops the brush pulling terrain below the grade.

Vegetation And Water

Press `V` for vegetation mode, `W` for water.

In vegetation mode, keys `0–7` select the preset to paint. Water mode paints the water mask — white is water, black is land.

Selection

Press `M` in edit mode.

Tool	Use
Rect / Lasso	Click and drag to select a region
Wand	Click to flood-fill select similar values

Key	Action
`Ctrl+C`	Copy
`Ctrl+V`	Paste — then click to place
`Ctrl+X`	Cut / export PNG
`Delete`	Clear the selected region

Copy and paste move terrain between locations, which is the quickest way to repeat a landform.

Tile Generation

Open the **Generate** panel to build terrain from real-world data.

Action	Result
Click a tile cell	Queue it
Drag across cells	Box-select a region to queue
RMB a cell	Remove from the queue
Scroll	Zoom the grid
MMB drag	Pan the grid
Run	Generate everything queued

Queue the whole area first, then Run once — generation is the slow part.

Tile Cleanup

A dedicated workspace for trimming large generated blocks down to what a railroad actually needs.

Action	Result
Drag	Mark tiles
`Shift`+drag	Add to the marked set
`Ctrl`/right-drag	Keep tiles

Action	Result
Invert / Outside ROW	Mark everything outside the right-of-way

The usual workflow is to mark the ROW, then **Invert / Outside ROW**, then delete.

Batch deletion asks for confirmation and moves files into a timestamped `_TileEditor_Deleted_Tiles` recovery folder. `Ctrl+Z` **restores the whole batch** — nothing is destroyed immediately.

OSM Guide Overlay

Press `O` for the OpenStreetMap overlay, which is the practical way to trace real alignments.

The in-game F9 equivalent streams an aligned guide over just the 5×5 or 8×8 game-tile neighbourhood around the camera, following terrain and unloading behind you. It shares the desktop editor's `Map.json` georeference and caches images under the installed mod rather than a machine-specific path.

Setting	Options
Mode	**Clear Lines** (default — drops the pale map sheet, keeps features) or **Full Map**
Resolution	Overview → Ultra, matching z15–z18

Adaptive detail keeps the camera area sharp without rendering the whole coverage window at maximum resolution. Both the desktop OSM toolbar and the F9 Terrain controls show cache usage and offer a confirmation-protected **Clear Cache**.

OSM and Mapbox data come from third-party services under their own terms — check those before redistributing anything derived from them.

In-Game Terrain Workspace

The F9 Terrain workspace separates practical sculpting from vegetation and water painting. Its building-pad, track/road, walkway, grade-plane, ditch, and embankment tools use bounded cut/fill with target convergence, specifically to avoid the spikes a naive flatten produces.

Saving

`Ctrl+S` saves all modified tiles. `Ctrl+X` exports the heightmap as a PNG. Press `D` for diff mode to see exactly which tiles are unsaved — they carry a yellow border.

Related

- Keybinds
- Getting Started
- Track Editing